

Software Engineer · AI Systems

Building AI technologies with AI tools.

I learned Computer Science first, then learned the inside of LLMs by hand — the substrate underneath and the tools on top. When an AI tool is wrong, I notice. When a model fails silently, I know how to diagnose it.

[See everything I've built >](#)

FOUNDATIONS

Hand-implemented transformer and GPT internals from scratch — **attention, backpropagation, tokenization, embeddings** — following Karpathy's **Neural Networks: Zero to Hero** end-to-end, before adopting AI-assisted development. Self-directed coursework in the math and ML underneath: **Stanford CS229**, MIT OpenCourseWare Linear Algebra, Differential Equations, Calculus, and MIT 6.034 Artificial Intelligence.

SELECTED PROJECTS

Rocket Engine Health Orchestrator

Python · PyTorch · Transformers · Three.js · 2026

Per-channel encoders fused by attention to catch **coupled, sub-threshold faults** that per-channel redlines are structurally blind to. Advisory above the deterministic redline, which keeps authority. Architecture, physics based data generator designed end to end. Design done, dev started; no model trained.

[Demo >](#) · [Case Study >](#) · [How It Works >](#)

Weather: Physics vs. ML

JavaScript · Python · ECMWF AIFS · BigQuery · 2026

A live evaluation system comparing GFS, ECMWF AIFS, and DeepMind WeatherNext, scored against Iowa Mesonet ASOS ground truth. **The real discipline was the verification loop** — caught two silent data-source bugs that would have corrupted the scoreboard.

[See the Case Study >](#)

Turbofan Predictive Maintenance

Python · PyTorch · Transformers · Flask · 2025–2026

Dual-head Transformer for jet-engine anomaly detection and Remaining Useful Life. **Diagnosed a total model collapse to a data signal-to-noise problem, not the model** — rebuilt the data, then derived a cost-weighted operating point cutting missed failures from 586 to 13 on the held-out test set.

[See the Case Study >](#)

Pong from Pixels

Python · PyTorch · Gymnasium · CUDA · 2026
Deep Q-network learning Pong from raw pixels on a 6 GB GPU. **Network, replay buffer, exploration schedule, and training loop hand-written**; emulator and autograd imported and named as such. Ongoing four-hour Saturdays, pre-registered floor and aspiration goals. Training not yet started.

[See the Case Study >](#)

AskMickey — AI Disney Assistant

Dart · Flutter · Gemini · iOS · Android · Web · Desktop · 2025

Designed a **classify-then-route dispatcher** on Gemini — MoE-inspired but deterministic — routing each query to one of eight handlers before any generative call, with **safety paths resolved at classification time** rather than left to the model. Shipped and live on four platforms from a single codebase.

[See the Case Study >](#)

Technical Skills

AI / ML — PyTorch · Transformers · LLMs / RAG · Agentic AI · Multi-Task Learning · scikit-learn
Languages — Python · C · C++ · Java · JavaScript · Dart · Swift · SQL
Frameworks — Flask · Flutter · Pandas · NumPy · Matplotlib
Infrastructure — Linux / UNIX · Amazon EC2 · Apache · MySQL
Tools — Git · VS Code · Arduino · Raspberry Pi
Practice — Feature branches · PRs carrying reasoning · Versioned releases

EXPERIENCE — EPCOM CORPORATION

Machine Learning Software Engineer Intern · Summers 2024–2025

Built a Transformer-based predictive-maintenance system for multivariate time-series sensor data. Integrated retrieval-augmented LLMs via prompt engineering and structured outputs; built data pipelines and evaluation harnesses for iterative development and benchmarking.

Software Development Intern · Summers 2020–2023

Full-stack components of an Applicant Tracking System (Linux · Apache · MySQL · PHP · JavaScript · Bootstrap) on Amazon EC2; automated cron jobs with pattern recognition to ingest applicants from IMAP email.

EDUCATION

University of Florida · August 2025

B.S. Computer Science

St. Petersburg College

A.A., Cum Laude

Volunteer Coach · Town of Belleair

Basketball (Coach of the Year) & Tennis, ages 4–16